Semantic HTML

A structured lesson on writing meaningful, accessible, and maintainable HTML.

Meaning

Elements that describe what content *is*, not just how it looks.

Accessibility

Screen readers and assistive tools rely on semantic structure.

SEO

Search engines rank pages better when structure is clear.

Maintainability

Developers read and update code faster with semantic markup.

What is Semantic HTML?

Semantic HTML uses HTML elements that clearly describe the meaning and purpose of their content. It helps browsers, developers, search engines, and screen readers understand the page structure.

Core Semantic Elements

- **<header>** — Header section
- **<nav>** — Navigation links
- **<main>** — Main content
- **<section>** — A thematic section
- **<article>** — Independent content
- **<footer>** — Footer information

Example

```
<header>
 <h1>My Website</h1>
</header>
<main>
 <section>
 <h2>About Us</h2>
 <p>Welcome to our website.</p>
 </section>
</main>
<footer>
 <p>
```

Why Use Semantic HTML?

Benefits of Semantic Elements

1 Readability

Code becomes easier to understand and maintain.

2 Accessibility

Screen readers can understand the structure of the webpage.

3 SEO

Search engines can better understand the importance and structure of content.

4 Maintainability

Developers can quickly identify what each part of the page represents.

Non-Semantic vs Semantic

```
<!-- Non-semantic -->
<div class="header">...</div>
<div class="navigation">...</div>

<!-- Semantic -->
<header>...</header>
<nav>...</nav>
```

Key Point: Semantic HTML describes what the content means, not just how it looks.

Important Semantic Layout Elements

Common Semantic Elements

- **<header>** — Introductory/header content
- **<nav>** — Navigation links
- **<main>** — Primary content
- **<section>** — Thematic grouping of content
- **<article>** — Independent/reusable content
- **<aside>** — Related or secondary content
- **<footer>** — Footer information

Typical Page Structure

```
<header>
 Logo + Website Title
</header>
<nav>
 Home | About | Contact
</nav>
<main>
 <section>
 Main Content
 </section>
 <aside>
 Related Content
 </aside>
</main>
<footer>
 Copyright Information
</footer>
```

Block & Layout Elements

A block-level element normally starts on a new line and occupies the available width of its container.

Examples of Block Elements

```
<div>Block Element</div>
<p>Paragraph</p>
<h1>Heading</h1>
<section>Section</section>
<header>Header</header>
<footer>Footer</footer>
```

Characteristics

- Starts on a new line by default.
- Usually takes the full available width.
- Can contain other elements depending on HTML rules.

Important: Block-level behavior is related to the element's default CSS display value. CSS can change how an element is displayed.

and Layout Structure

The `<div>` element is a generic container used to group content. It carries no semantic meaning on its own.

When Should We Use ?

Use `<div>` when:

- No suitable semantic element exists.
- You need a container for CSS styling.
- You need to group elements for JavaScript.

vs Semantic Elements

```
<!-- Generic -->
<div class="article">
  <h2>HTML</h2>
  <p>Learn HTML...</p>
</div>

<!-- Semantic -->
<article>
  <h2>HTML</h2>
  <p>Learn HTML...</p>
</article>
```

Rule: Prefer semantic elements when they accurately describe the content.

<iframe>

The `<iframe>` element stands for **Inline Frame**. It allows another HTML document or supported external content to be embedded inside the current webpage.

Syntax

```
<iframe
  src="https://example.com"
  width="600"
  height="400">
</iframe>
```

Best Practice

```
<iframe
  src="..."
  title="Google Map showing college location"
  loading="lazy">
</iframe>
```

Common Uses

- Embedding maps
- Embedding videos
- Embedding documents
- Displaying another webpage
- Integrating external content

Important Attributes

- **src** — Specifies the content URL
- **width** — Sets width
- **height** — Sets height
- **title** — Describes the iframe for accessibility
- **loading** — Can control loading behavior

<head> Element

The <head> contains metadata and resources about the webpage. This information generally isn't displayed as the main page content.

Common Elements Inside <head>

```
<head>
  <meta charset="UTF-8">
  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0">
  <title>My Website</title>
  <link rel="stylesheet" href="style.css">
  <script src="script.js" defer></script>
</head>
```

<head> vs <body>

<head>

- Metadata
- Title
- CSS references
- Scripts
- SEO-related information

<body>

- Headings
- Paragraphs
- Images
- Forms
- Tables
- Visible page content

Meta Tags

`<meta>` elements provide metadata about the webpage. They are placed inside the `<head>`.

Character Encoding

```
<meta charset="UTF-8">
```

Tells the browser how text should be encoded.

Responsive Viewport

```
<meta
  name="viewport"
  content="width=device-width,
initial-scale=1.0">
```

Helps webpages display correctly on different screen sizes.

Description

```
<meta
  name="description"
  content="Learn HTML, CSS and
JavaScript.">
```

Provides a description of the webpage that search engines may use in search results.

Important Meta Tags

Commonly Used Meta Tags

```
<meta charset="UTF-8">
<meta
  name="viewport"
  content="width=device-width, initial-scale=1.0">
<meta
  name="description"
  content="HTML learning website">
<meta
  name="author"
  content="Web Development Team">
```

What They Do

- **charset** — Defines character encoding
- **viewport** — Controls mobile viewport behavior
- **description** — Describes page content
- **author** — Identifies the content author

Important Note: Meta tags don't replace good page content or semantic HTML. They provide additional information about the document.

Complete HTML Structure + Quick Recap

Putting Everything Together

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta
    name="viewport"
    content="width=device-width, initial-scale=1.0">
  <meta
    name="description"
    content="HTML Learning Website">
  <title>HTML Learning</title>
</head>
<body>
  <header>
    <h1>HTML Learning</h1>
  </header>
  <nav>
    <a href="#">Home</a>
    <a href="#">Topics</a>
  </nav>
  <main>
    <section>
      <h2>Semantic HTML</h2>
      <p>Semantic elements describe the meaning of
content.</p>
    </section>
    <aside>
      Related Topics
    </aside>
  </main>
</body>
</html>
```

Final Recap

- **Semantic HTML** → Gives meaning to page structure.
- **Block elements** → Usually begin on a new line and occupy available width.
- ** ** → Generic grouping/container element.
- **<iframe>** → Embeds external content.
- **<head>** → Contains document metadata and resources.
- **<meta>** → Provides metadata such as encoding, viewport, and description.

🎯 **Lab Task:** Create a complete webpage using: `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`, `<iframe>`, `<title>`, and at least 3 different `<meta>` elements.